



Additionally, SiB can automatically connect to a secondary Wi-Fi access point (once configured) if the

primary network is down. In other words, SiB keeps running even during power or network interruptions. The

device is available for pre-order on Kickstarter, and is available at a starting price of US\$ 5.

4 DRONE THAT CAN FLY On Its Own

Unmanned aerial vehicles (UAVs) are reaching new heights of advancements as underlying components and programs keep improving. Drones—one of the most popular forms of UAVs today—are being used in a variety of applications, including personal use, package delivery, professional photography and videography, and surveillance. Beijing-headquartered ZeroZero Robotics has taken drones a step further, by designing an autonomous drone called Hover 2.

Hover 2 is a compact, intuitive and interactive UAV that allows users to get high-quality photos and videos from any angle and height achievable by a drone. In its inactive form, it can be folded into a box form factor, jam-packed with numerous elements.

To fly it, user need to simply open the folds to reveal a caged casing containing the propellers, attached to the central body that homes the various components. Power button on the body powers up the drone. An accompanying mobile application and a remote is needed to configure and command the drone. Once configured, the drone can be placed mid-air and it will take off on its own!

The 500-gram object has various elements, including 360-degree optical radar, two-axes mechanical gimbal with electronic image stabilisation, in-built 12MP camera lens with 78.8-degree field-of-view, up to 4K resolution video and photo compatibility, and Qualcomm processor for computation.

Hover 2 features artificial intelligence (AI)-enabled image capture capabilities, including capturing by auto-frame, taking track shots,



cinematic theme images and omnidirectional capture (that is, from the front, sides and back).

A single-touch command on the app sets the drone in facial recognition mode and allows it to capture videos and photos, revolving around the users in orbital, ricochet or track modes while detecting their face. This facial recognition feature, along with target lock algorithms, can also be used to make the drone follow someone. The drone creates a digital model of its environment to identify any obstructions while staying locked on to the target, ensuring it keeps following the intended person, even amidst distractions.

The optical radar remains contracted inside the drone body. When powered on, it protrudes out and swivels 360 degrees along its axis to pick up objects in proximity. The drone's processor runs proprietary algorithms

of SLAM 3D mapping, visual inertial odometry and path planning, along with readings from the radar to avoid any kind of obstacle and plan its flight route on-the-go. Onboard GPS keeps a tab on the navigation route and helps the drone return to base safely. The drone is powered by a lithium-ion battery that can keep it in the air for up to 23 minutes at a go.

The drone can also be availed with BlastOff feature, which requires a set of enhanced propellers and an exclusive controller. This enables it to reach greater heights; it can be controlled from a distance of up to five kilometres.

Hover 2, built by Stanford PhDs Tony Zhang and MQ Wang, is available online, at a starting price of US\$ 399. The basic box set comes with the device, charger, propeller guards, remote controller, landing gear and battery. **EFY**